

WEEK 08 · JAVASCRIPT · THE AI WORKFLOW

THE LANGUAGE YOU ALREADY KNOW

JavaScript for people who already program — and the method you will use for the rest of the course.

WHERE WE ARE

Six weeks of pages that render and work. Today they start to think.

AFTER TODAY YOU CAN

Read JavaScript that surprises you, and say why it did.

WHAT THIS UNLOCKS

Your three pages, and the first spec you write before the code.

“Java and JavaScript are as similar as car and carpet.”

THE OLD COURSE’S JAVASCRIPT LESSON, KEPT — STILL THE MOST USEFUL SENTENCE IN THIS DECK

TARGET · MINUTE 0

THIS IS WHERE YOU WILL BE AT MINUTE 180

the bench of class assignment 8, every row green – four pure functions and one repaired review

reload

open

The bench

Parts 1 and 2. Save a file, refresh this page. Green here is the floor, not the grade.

FILTERBYQUERY

5 of 5 passing

finds one, ignoring case

an empty query returns everything

spaces around the query are ignored

Hebrew works too

it does not touch the array it was given

SORTITEMS

4 of 4 passing

The bench

Parts 1 and 2. Save a file, refresh this page. Green here is the floor, not the grade.

FILTERBYQUERY

5 of 5 passing

finds one, ignoring case

an empty query returns everything

spaces around the query are ignored

Hebrew works too

it does not touch the array it was given

SORTITEMS

4 of 4 passing

by title, and returns to default property

by year, added first, not year last

by rating, then reviews by title

one property, is NOT recommended

targets/week-08.png · 1280

סיכום פריט [הרשימה](#) מדף התקליטים

המדף

התקליטים שאספתי, עם דירוג ושנה.

375 · the list page, toolbar outside the list

► **THE POINT** — four things leave this room: fourteen predictions with the gap explained, four **PURE** functions that pass this bench without touching the array they are handed, one review of code Claude wrote, and your first `FEATURE_SPEC.md` with three pages built from it. You do each once in a cycle first.

AI THIS WEEK

PARTNER. IT WRITES THE CODE; YOU WROTE THE SPEC, AND YOU CHECK WHAT CAME BACK.

LAYER · WEEKS 8–12 · PARTNER

ALLOWED

Claude implements a feature **from your spec**: the filter and sort in cycles 4–5, the three pages of part 4 from your `FEATURE_SPEC.md`. Asking it to review — as a claim, checked against the bench.

NOT ALLOWED

Code with no spec. A commit with no review line. And parts 1–3 of the assignment — predict, the four functions, the review — **you read and write those alone**, because week 13 does.

NO SUBSCRIPTION?

Everything works in the free chat, and nothing in the course depends on a paid plan. Paste the spec, copy the file back by hand, run the same bench. Graded identically.

WHAT TODAY IS, AND IS NOT

NOT TODAY

Variables. Loops. `if`. Functions as an idea. Arrays as an idea. You have all of it.

TODAY

The seven places where what you know is **wrong here**, and the idioms your project is built from.

AFTER THE BREAK

The method: a spec to decide, Claude to write, a bench to check, and a review that is graded.

► **SAY THE DEAL OUT LOUD** — if any of this is genuinely new rather than merely different, `primer/` covers it, with solutions, and nobody is graded on it. Hoisting, the temporal dead zone, closures and the event loop are in the reference deck, not in a cycle — week 10 and week 11 use them.

CYCLE 1 · 25 MIN

COERCION AND TRUTH: WHAT THE FORM GAVE ME

A form field gives you a string, or nothing at all. One function, `stockLabel(row)`, turns what arrived into a label — and on the way: the `==` triangle, why `===`, the one exception (`== null`), `Number('')` that is zero, and `??` that keeps a zero a zero. `app.js` is a module: Live Server, and `import` with `./` and `.js`.

1 I DO

10 MIN

3 on the next three slides, 7 in the editor · live-code/01-coercion-and-truth · the triangle · if (`row == 0`) (wrong) · trim, `Number`, `Number.isNaN`, `===` · (`row ?? ''`) · copies `||` / `??`

2 WE DO

5 MIN

you type steps 1, 3, 4 · compare with checkpoint 1 · fourteen rows, "not entered" twice

3 YOU DO

8 MIN

`priceLabel(row): "no price" · "free" · "12.5" · "not a number" — and no == inside it`

4 DEBRIEF

2 MIN

`Number(row)` before the empty check makes the empty field "free" · ↗ reference

SIX LINES. EVERY ONE OF THEM IS LEGAL.

```
1 [] + []           // ""
2 [] + {}           // "[object Object]"
3 0.1 + 0.2          // 0.30000000000000004
4 [10, 9, 1].sort() // [1, 10, 9]
5 typeof null        // "object"
6 '5' - 3            // 2      ... but '5' + 3 is "53"
```

You will not need **any** of these lines. You will need to not be surprised by them.

THE TABLE NOBODY MEMORISES, COMPUTED LIVE

every cell is a real `a == b`, and it explains itself

↺ reload

open ↗

LOOSE EQUALITY, EVALUATED BY THIS PAGE

☒ compare with == ☐ compare with === ☐ highlight the broken triangle

Click a cell to see how the specification got there.

==	0	1	'0'	''	'false'	false	true	null	undefined	NaN	[]	[0]
0	true	false	true	true	false	true	false	false	false	false	true	true
1	false	true	false	false	false	false	true	false	false	false	false	false
'0'	true	false	true	false	false	true	false	false	false	false	false	true
''	true	false	false	true	false	true	false	false	false	false	true	false
'false'	false	false	false	false	true	false	false	false	false	false	false	false
false	true	false	true	true	false	true	false	false	false	false	true	true
true	false	true	false	false	false	false	true	false	false	false	false	false
null	false	false	false	false	false	false	false	true	true	false	false	false
undefined	false	false	false	false	false	false	false	true	true	false	false	false
NaN	false	false	false	false	false	false	false	false	false	false	false	false
[]	true	false	false	true	false	true	false	false	false	false	false	false
[0]	true	false	true	false	false	true	false	false	false	false	false	false

35 of 144 pairs are equal under ==. The diagonal accounts for 9 of them. Everything else is a conversion you did not ask for.

HOW IT GOT THERE

Pick a cell.

► ASK FOR `'' == 0`, THEN `0 == '0'`, THEN `'' == '0'` — and wait. Only then switch on the broken triangle.

THE RULE, AND THE ONE EXCEPTION WORTH KEEPING

NEVER

```
1 if (value == 0)
2 if (input == false)
3 // "" passes. [] passes.
```

ALWAYS

```
1 if (value === 0)
2 if (input === false)
```

THE EXCEPTION

```
1 if (value == null)
2 // null AND undefined,
3 // and nothing else
```

=== asks one question. **==** asks a question and then negotiates.

CHECKPOINT 1

YOUR SCREEN SHOULD LOOK LIKE THIS

Week 8 · cycle 1 · coercion and truth

What the form gave me

One row for every print() in app.js, in the order they ran.

what app.js printed	the value
<code>'' == 0</code>	true
<code>0 == '0'</code>	true
<code>'' == '0'</code>	false
<code>'' === 0</code>	false
<code>Number('')</code>	0
<code>stockLabel("3")</code>	"3 in stock"
<code>stockLabel("0")</code>	"sold out"
<code>stockLabel("")</code>	"not entered"
<code>stockLabel(" ")</code>	"not entered"
<code>stockLabel("abc")</code>	"not a number"
<code>stockLabel(null)</code>	"not entered"
<code>copies 'unknown'</code>	"unknown"
<code>copies ?? 'unknown'</code>	0
<code>null == undefined</code>	true

Compare — `live-code/01-coercion-and-truth/checkpoint.png`. Fourteen rows · `stockLabel("")` and `stockLabel(" ")` say "not entered", not "sold out" · `stockLabel(null)` prints a row and the console is clean · `copies ?? 'unknown'` is 0.

CYCLE 2 · 25 MIN

THE ARRAY YOU WERE HANDED

A function that sorts the list it receives — without touching it. `sort` reorders in place **and** returns the array, so it looks safe; `[...list]` before every `sort`; a comparator returns a number; a missing year goes last with `?? Infinity`; Hebrew and English sort with `localeCompare(..., 'he')`.

1 I DO

10 MIN

3 on the next four slides, 7 in the editor · live-code/02-the-array-you-were-handed · `list.sort(...)` (wrong) · `[...list]` · `null - 1815` · `?? Infinity` · `[...names].sort()` vs `localeCompare`

2 WE DO

5 MIN

you type steps 2–4 · compare with checkpoint 2 · the first row and the last row are the same

3 YOU DO

8 MIN

`newestFirst(list)` — newest first, the book with no year still last · AI as partner: ask why, then print it

4 DEBRIEF

2 MIN

the function returned the right answer — what was wrong? · ↗ reference

YOU DO · AI

➤ REFERENCE · 14, 15, 17–21

SIX PROGRAMS, AND THE ENGINE ANSWERS

identical steps, one difference – and an `a === b` asked after each

⌕ reload open ↗

SIX PROGRAMS, STEPPED, WITH THE ENGINE ANSWERING

Program 1 · a number is copied ▼ Step Reset step 0 of 3

```
let a = 10;  
let b = a;  
b = 99;
```

Press Step.

BINDINGS, AFTER THIS STEP

NAME	VALUE
a	undefined
b	undefined

► **STOP PROGRAM 4 ON ITS LAST STEP.** Ask who thinks `prices` is unchanged. Count the hands. Then press Step.

THE FUNCTION THAT TOLD THE TRUTH AND LIED

WHAT IT SAYS

```
1 function addTax(list) {  
2   list.sort((a, b) => a - b);  
3   return list.map((p) => p * 1.17);  
4 }
```

Returns a new array. **And reorders yours.**

WHAT IT SHOULD SAY

```
1 function addTax(list) {  
2   const copy = [...list];  
3   copy.sort((a, b) => a - b);  
4   return copy.map((p) => p * 1.17);  
5 }
```

One line. Now the name is honest.

sort and **reverse** mutate **and**
return the array, so they look safe.

const IS ABOUT THE NAME

```
1 const list = [1, 2];
2 list.push(3);           // fine. the array was never frozen.
3
4 const total = 0;
5 total = 1;              // TypeError: Assignment to constant variable.
6
7 // use const everywhere. use let when the NAME must point somewhere else.
```

THE HABIT

`const` by default. `let` when you reassign. `var` never — the reference deck says why, and the assignment checks it.

NOT A PROMISE OF IMMUTABILITY

If you actually need that, copy: `[...list]`, `{ ...item }`. Both are shallow.

COMPARATORS, AND THE ONLY TWO YOU NEED

WRONG

```
1 list.sort()
2 list.sort((a, b) => a.title > b.title)
```

The first compares strings. The second returns `true / false`, and `sort` wants a **number**.

RIGHT

```
1 [...list].sort((a, b) => a.year - b.year)
2 [...list].sort((a, b) =>
3   a.title.localeCompare(b.title, 'he'))
```

Negative, zero, positive. And a copy, so the caller keeps its order.

A MISSING VALUE GOES LAST

```
1 (a, b) => (a.year ?? Infinity)
2           - (b.year ?? Infinity)
```

`null - 1815` is `-1815`: without `??` the book with no year comes **first**. `?? 0` does the same. Every real year is smaller than `Infinity`.

TIES ARE NOT FREE

```
1 b.rating - a.rating
2 || a.title.localeCompare(b.title, 'he')
```

CHECKPOINT 2

YOUR SCREEN SHOULD LOOK LIKE THIS

Week 8 · cycle 2 · the array you were handed

The list you were handed

One row for every print() in app.js, in the order they ran.

what app.js printed	the value
shelf, as handed	[{"The Hobbit (1937)", "(1968) 'מִיכָאֵל שֵׁלִי", "dune (1965)", "Field notes (?)", "Emma (1815)"}]
null - 1815	-1815
byYear(shelf)	[{"Emma (1815)", "The Hobbit (1937)", "dune (1965)", "(1968) 'מִיכָאֵל שֵׁלִי", "Field notes (?)"}]
[...names].sort()	[{"Emma", "Field notes", "The Hobbit", "dune", "'מִיכָאֵל שֵׁלִי"}]
byTitle(shelf)	[{"(1968) 'מִיכָאֵל שֵׁלִי", "dune (1965)", "Emma (1815)", "Field notes (?)", "The Hobbit (1937)"}]
search(shelf, ' HO ')	[{"The Hobbit (1937)"}]
countByDecade(shelf)	[{"label": "1960s", "count": 2}, {"label": "?", "count": 1}, {"label": "1810s", "count": 1}, {"label": "1930s", "count": 1}]
mean · Math.round · toFixed	[1921.25, 1921.3, "1921.3"]
shelf, afterwards	[{"The Hobbit (1937)", "(1968) 'מִיכָאֵל שֵׁלִי", "dune (1965)", "Field notes (?)", "Emma (1815)"}]

Compare — `live-code/02-the-array-you-were-handed/checkpoint.png` . The first row and the last row are identical · `Field notes (?)` is last in `byYear` · `byTitle` starts with the Hebrew title · `toFixed(1)` is in quotes.

CYCLE 3 · 30 MIN

THIS AND ARROWS: WHOSE `this` IS IT?

Functions are values — you hand them over by name, without parentheses. An arrow with braces needs a `return`. And `this` is decided where the function is **called**, not where it was written. The rule that comes out of it: **arrow for callbacks, function for methods**.

1 I DO

10 MIN

3 on the next three slides, 7 in the editor · live-code/03-this-and-arrows · `(book) => { book.title }` (wrong) · `map(toTitle)` vs `map(toTitle())` · `count()` and `count`, detached · `tagged()` with an arrow inside

2 WE DO

8 MIN

you type steps 2–4 · `break` tagged() once with function (`book`), then put the arrow back · compare with checkpoint 3

3 YOU DO

10 MIN

lab/warmup/ — five functions, four wrong, the referee says "five of five"

4 DEBRIEF

2 MIN

"arrows fix this, so everything is an arrow" breaks the method · ↗ reference

YOU DO · AI-FREE

➤ REFERENCE · 24, 25

ARROW OR FUNCTION – AND IT IS NOT ABOUT BREVITY

SAME

```
1 const a = (x) => x * 2;  
2 function b(x) { return x * 2; }
```

THE TRAP

```
1 const f = (x) => { x * 2 };  
2 f(3); // undefined
```

Braces make a **body**. A body needs `return`.

EITHER OF THESE

```
1 const f = (x) => x * 2;  
2 const g = (x) => {  
3   return x * 2;  
4 };
```

Returning an object needs the parentheses:

`(item) => ({ id: item.id })`.

Without them the braces are a body again.

ONE FUNCTION, FIVE CALL SITES

nothing is redefined between the rows – only the call changes

↺ reload

open ↗

ONE FUNCTION OBJECT, FIVE CALL SITES

```
const owner = { name: 'the shelf' };

function probe() {
  return this?.name ?? '(no this)';
}

owner.probe = probe;           // the SAME function, given a second name
```

Run all five

☐ define `probe` as an arrow instead

HOW IT IS CALLED	THIS?.NAME	WHY
<code>probe()</code>	(no this)	Nothing before the dot, because there is no dot. In a module (and in strict mode) <code>this</code> is <code>undefined</code> rather than the window.
<code>owner.probe()</code>	the shelf	The object before the dot becomes <code>this</code> . This is the rule people think is the only rule.
<code>probe.call(other)</code>	a different object	<code>'call'</code> , <code>'apply'</code> and <code>'bind'</code> say what <code>this</code> is, out loud. This is the escape hatch.
<code>[1].map(owner.probe)[0]</code>	(no this)	The function was passed, not called through the object. <code>'map'</code> calls it with no object in front, so the object is gone. This is the bug you will hit in week 9.
<code>[1].map(() => owner.probe())[0]</code>	the shelf	The arrow is the callback; the call inside it is still a method call through <code>'owner'</code> . The fix is not "use an arrow" — it is "keep the dot".

`probe === owner.probe` is true. There is one function. It does not belong to `'owner'`, it is not a copy, and nothing about it changed between the rows above — only the call site did.

► **READ THE FIVE CALL SITES ALOUD AND VOTE ON EACH, BEFORE REVEALING.** The method row is unanimous. The callback row is not.

FOUR RULES, CHECKED IN THIS ORDER

- 1 new**
a brand-new object. Not used in this course.
- 2 call / apply / bind**
whatever you said, explicitly.
- 3 a dot**
`owner.probe()` — the thing before the dot.
- 4 none of the above**
`undefined` in a module. Not the window.

AND OUTSIDE ALL FOUR

An **arrow** has no `this` of its own, so none of the rules apply. It uses the `this` of wherever it was written.

THE PRACTICAL RULE

Arrow for callbacks. Function for methods. That is the whole decision.

Also: `this?.label` is `undefined`, not a crash, when `this` is `undefined`.

CHECKPOINT 3

YOUR SCREEN SHOULD LOOK LIKE THIS

Week 8 · cycle 3 · this and arrows

Whose this is it?

One row for every print() or attempt() in app.js, in the order they ran.

what app.js printed	the value
shelf.books.map(toTitle)	["The Hobbit", "dune", "היכל של"]
shelf.books.map(toTitle())	TypeError: Cannot read properties of undefined (reading 'title')
shelf.count()	"Reading list: 3 books"
count(), no dot	TypeError: Cannot read properties of undefined (reading 'label')
shelf.tagged()	["Reading list · The Hobbit", "Reading list · dune", "Reading list · היכל של"]

Compare — `live-code/03-this-and-arrows/checkpoint.png`. Five rows, two of them red **on purpose**: `map(toTitle())` and `count(), no dot`. They are the proof of the two rules, and they stay.

BREAK

10 MINUTES

Back at 100. Cycles 4 and 5 are the AI cycles — a chat window open (free or paid), `project/templates/FEATURE_SPEC.md` and `PROMPTS.md` where you can copy them, and a copy of `ai-lab/04-spec-then-generate/artifact/` as your practice folder.

CYCLE 4 · 25 MIN

A SPEC FIRST, THEN CLAUDE WRITES

You copy the template into `specs/filter-and-sort.md` and fill it — \$2 says `books` is never reordered. Claude implements it. The answer reads well, line by line, and the bench finds two red rows: the right list returned, **your** list reordered. Then you write the prompt again, with the rule in it.

1 I DO

8 MIN

3 on the next three slides, 5 in the editor and the chat · ai-lab/04-spec-then-generate · the good spec on screen · specs/filter-and-sort.md typed · P1 live

2 WE DO

6 MIN

your own specs/filter-and-sort.md from the template · P1 in your chat · bench.html · find the red row before it gets a name

3 YOU DO

9 MIN

your own P2 with the rule in the prompt · 9 of 9 · search the file for document, \$, var, export

4 DEBRIEF

2 MIN

a rule in the spec is not a rule that is kept · ↗ reference

YOU DO · AI

THE LOOP

Claude chat

- 1 to think. Ask for options.

No code yet.

- 2 **FEATURE_SPEC.md**

to decide. Written down is checkable.

- 3 **Claude Code**

to implement — from the spec, not from a sentence.

- 4 **Review**

against your own criteria. `git diff`, not memory.

- 5 **Iterate the spec**

, not the output. **Nobody does step 5.** It is the one that compounds.

THE SAME FEATURE, FROM TWO REQUESTS

both implementations are real, and both are run against the same eight cases

↺ reload

open ↗

ONE FEATURE, TWO REQUESTS, EIGHT CASES

☐ only run the happy path Click a row to see which line of the spec it comes from.

THE REQUEST MOST PEOPLE SEND

"Add a search box that filters the list."

1. — nothing else was said —

```
export function filterItems(items, query) {  
  return items.filter((item) =>  
    item.title.includes(query)  
  );  
}
```

fails 4 of 8

THE SAME FEATURE, SPECIFIED

"Implement `filterItems` per the spec below."

1. Section 2 — `query` is a string on state; it starts as `''`.
2. Section 5 — an empty query shows everything, not nothing.
3. Section 5 — leading and trailing spaces are trimmed before matching.
4. Section 5 — matching is case-insensitive.
5. Section 5 — an item whose title is missing must not crash the list.
6. Section 6 — with no matches the list is empty and the counter reads 0.

```
export function filterItems(items, query) {  
  const term = query.trim().toLowerCase();  
  if (term === '') return items;  
  return items.filter((item) =>  
    (item.title ?? '').toLowerCase().includes(term)  
  );  
}
```

passes all 8

► **TICK "ONLY RUN THE HAPPY PATH" FIRST.** Both green, both finished-looking. Then untick it, and say nothing.

WHAT A SPEC HAS THAT A SENTENCE DOES NOT

- §1 What the user can do after, and could not before. One sentence. If it contains "and", it is two features.
- §2 What changes in `state` — before and after.
This week that is your data model.
- §5 Edge cases, at least three. **AI writes the happy path very well and the edges almost never.**
- §6 How you know it works — things you can **see**, not "the function returns an array".
- §7 The review, written **after** the code exists.
This is the graded part.

YOU ALREADY HAVE THE FILE

`project/templates/FEATURE_SPEC.md` . Copy it once per feature into `specs/` — today, `specs/filter-and-sort.md` .

THIS WEEK ONLY

§4 (the events) says “**not yet — week 9**”, in Hebrew in your file. There is no behaviour yet, and pretending otherwise is worse than an empty section.

CYCLE 4 · ON SCREEN BEFORE ANYONE WRITES

WHAT A SPEC THAT WORKS LOOKS LIKE

§2 · WHAT IS NOT ADDED

`visibleShelf` is **not** stored — the list on screen is derived. That line is what Claude will try to add, so it is written down before it asks.

§5 · CASES THAT PASS THE DEMO AND FAIL A USER

Spaces only · no match · a year that is missing, **last, not year 0** · Hebrew and English together. Every red row on a bench is a line here, or should have been.

§7 · WRITTEN AFTER THE CODE

What came back, what was wrong, what I did. It is the graded part, and it is empty until cycle 5.

► **AI-EXEMPLARS/EXEMPLARS/FEATURE-SPEC-GOOD.MD** in your course materials — the short version of the project example's real spec, `project/templates/examples/FEATURE_SPEC.md` . Read it for §2, §5 and §7.

CHECKPOINT 4 · THREE QUESTIONS

BEFORE YOU LEAVE CYCLE 4

- 1 Did you find a row where the returned list is right and your list was reordered?**
And read it back to \$2 of the spec, before anybody named it.
- 2 Is the rule in the body of your P2, and does what came back pass 9 of 9?**
“Copy before any sort, on every path, including the empty query.”
- 3 Did you search the P1 file for what you did not ask for?**
`wireUp` and `$` — and `var` , found zero times. Write down the search that found nothing, too.

CYCLE 5 · 20 MIN

THE REVIEW, THEN THE JOURNAL

Claude is asked to review its own file against the spec. Seven rows, seven ticks — and the tick on \$2 repeats the sentence that wrote the bug. The bench says the opposite. Then you write the two documents that are graded: \$7 of the spec, and entry 1 of **PROMPTS.md** — from what ran, not from what was said.

1 I DO

7 MIN

2 on the next slide, 5 in the chat ·
ai-lab/05-review-then-journal · P3:
"review your file" · the bench
beside it · P4: the red row as
evidence

2 WE DO

5 MIN

the good review on screen, then
the bad one · \$7 of specs/filter-and-
sort.md, three rows, together

3 YOU DO

6 MIN

the good journal entry on screen,
then the bad one · entry 1 of your
project/PROMPTS.md

4 DEBRIEF

2 MIN

a review by the author is a claim;
the proof is what ran · ↗ reference

YOU DO · AI

ONE WINDOW, ONE FEATURE, ONE REVIEW

1 Left

the chat, or Claude Code — the file it wrote, and its own review of it.

2 Middle

`specs/filter-and-sort.md`, §7 still empty.

3 Right

the bench, `bench.html` — what ran, row by row.

4 Below

`git diff`: the thing nobody asked for (`wireUp`, `$`).

5 Then write

§7 of the spec, and entry 1 of `PROMPTS.md` — from the bench, not from the review.

► **A REVIEW BY THE AUTHOR IS A CLAIM.** The bench is the proof. **NO SUBSCRIPTION?** Paste the file and the spec into the free chat, then the red row — the review you write is identical, and it is the part worth points.

CYCLE 5 · ON SCREEN BEFORE PART 3 IS GRADED

A REVIEW OF GENERATED CODE: GOOD, THEN BAD

AI-REVIEW-JS-GOOD.MD

Which one — `byDecade`. **The claim, and the part of it that is true** — `Math.floor` is right, `=== undefined` does skip a missing field. **What happens** — `year: null` lands in decade 0. **The proof** — three lines and what they printed:
`{"0": ["Debut"], "2000": ["Kid A"]}`. **The fix** — `== null`, one change.

AI-REVIEW-JS-BAD.MD

Names the wrong function (“a Set is a hash, the order is random” — it keeps insertion order). “The explanation lied.” “I read it and understood.” Rewrote all three. **True in places, and worth nothing.**

► **AI-EXEMPLARS/EXEMPLARS/AI-REVIEW-JS-GOOD.MD** and **-BAD.MD** in your course materials — the same six sections as part 3 of the assignment, on three different functions.

CYCLE 5 · ON SCREEN BEFORE YOU WRITE ENTRY 1

A JOURNAL ENTRY: GOOD, THEN BAD

PROMPT-JOURNAL-GOOD.MD

The exact prompt. What came back. **What was wrong, and how I checked** — a click, a number, a bench row. What I did. And the line that turns one entry into a rule: **what I learned about the prompt itself** — “quote the rule in the body of the prompt”.

PROMPT-JOURNAL-BAD.MD

“I asked Claude to build the filter and sort.” “There were a few small things; I fixed them.” “Claude Code saves a lot of time.” **Every word is true, and it is worth nothing** — to the grader, and to you in week 13.

► **AI-EXEMPLARS/EXEMPLARS/PROMPT-JOURNAL-GOOD.MD** and **-BAD.MD**. Your entry 1 goes in `project/PROMPTS.md` (copy `project/templates/PROMPTS.md` if you have none) and it is about cycle 4: P1, the red row, P2, and what you learned about the prompt.

CHECKPOINT 5 · THREE QUESTIONS

BEFORE YOU LEAVE CYCLE 5

- 1 Does §7 of your spec quote the bench row, and not the tick from P3?**
The input, and the word REORDERED.
- 2 In the §2 row, did you write which part of the explanation is true, and why the conclusion is not?**
`filter` does return a new array. On the empty path there is no `filter`.
- 3 Is the last line of your journal entry a rule for the next prompt?**
Not “Claude is really good”.

ASSIGNMENT · 145-175

CLASS ASSIGNMENT 8: ASSEMBLE, THEN EXTEND

1 Predict · cycles 1-3

`predict-he.md` — fourteen snippets: guess, run, then say why. The why is graded, not the guess.

Four pure functions · cycles 1-2

2 `filterByQuery`, `sortItems` (a copy · ?? `Infinity` · `localeCompare`), `countByArtist` (count into an object), `summarise` (`Math.round(x * 10) / 10`). No `var`.

3 Review what Claude wrote · cycle 5

`topRated`, `initials`, `formatRating`: one does not do what its explanation says. Name it, prove it by running it, fix only it.

Spec, then pages · cycle 4

4 `FEATURE_SPEC.md` for your topic, then three pages: the toolbar *outside* `<ul id="list">`, `aria-current="page"` on the current link.

PARTS 1-3 ARE YOURS

No AI in predict, the functions or the review. **Week 13 reads code the same way, offline.**

PART 4 IS THE LOOP

Spec first, then Claude, then the review in §7. A spec written after the code scores zero there.

THE MILESTONE, AND HOMEWORK 3

THE PROJECT, BY THE END OF THIS WEEK

Three pages in HTML and Tailwind, invented data in the markup, a menu between them with `aria-current="page"`, the toolbar outside the list. **No JavaScript in the project yet** — week 9 wires it, week 10 draws it from state.

HOMEWORK 3 — ASSIGNED TODAY

Seven drum pads and two dice, **built entirely by JavaScript from two arrays**: one listener for every pad, a history you can remove from, a form that fails well. It is DOM work — week 9 is the week that teaches it. **Due before lesson 10.**

CLASS ASSIGNMENT 8

Moodle, repo URL + commit SHA, **before lesson 9**. The work sits in `week-08/`.

THREE QUESTIONS

- 1 **Why is `'' == '0'` false**
when `'' == 0` and `0 == '0'` are both true?
- 2 **A function returned the right sorted list.**
What else do you check before you call it correct?
- 3 **"`filter` returns a new array, so your list is safe."**
True sentence. When is the conclusion false?

If you can answer these three,
you can read the code of week 9.
If not, `live-code/*/end/` and
the demos are yours.

GO DEEPER

The reference deck for this week — every slide the lecture pointed at, in order.

[2 · Seven types, and one of them is a mistake](#) ← spine 5

[3 · Eight values are false. Everything else is not.](#) ← spine 5

[5 · ==, in the order the rules are applied](#) ← spine 5

[6 · Two ways to have nothing, and one to have a wrong number](#) ← spine 5

[7 · The two nothings, in arithmetic](#) ← spine 5

[8 · Converting on purpose](#) ← spine 5

[10 · Two operators that replace ten lines of guarding](#) ← spine 5

[12 · Modules: every file has a door](#) ← spine 5

[14 · You know this one. It is not the same here.](#) ← spine 10

[15 · Six methods, and you already know five of them](#) ← spine 10

[17 · The one that is not like the others](#) ← spine 10

[18 · Comparators](#) ← spine 10

[19 · toSorted\(\): the copy, built in](#) ← spine 10

[20 · sort is stable — and you still write the tie-breaker](#) ← spine 10

[21 · Taking things apart, and putting them together](#) ← spine 10

[23 · var, let, const](#) ← spine 13

[24 · Functions are values. That is not a slogan.](#) ← spine 16

[25 · this, by call site](#) ← spine 16

[27 · Three ways to declare, and they fail differently](#) ← spine 4

[28 · Hoisted is not the same as usable](#) ← spine 4

[29 · A closure is a function that remembers where it was born](#) ← spine 4

[30 · The same closure, in the file you will ship](#) ← spine 4

[31 · Build the store, in nine lines](#) ← spine 4

[32 · Where your code runs now](#) ← spine 4

[33 · What does this print?](#) ← spine 4

[34 · One stack, two queues, one rule](#) ← spine 4

[35 · The rule, in one sentence](#) ← spine 4

[36 · setTimeout\(fn, 0\) means "when I am free"](#) ← spine 4

[38 · From week 8, Claude writes code for you](#) ← spine 22

[39 · A spec you can check before you read any code](#) ← spine 22

[42 · The five that cost marks](#) ← spine 35

[43 · Where to look, in order](#) ← spine 35